

Javier Antonio Salazar Cavazos

Ann Arbor, Michigan, USA | javieresc1.github.io | javieresc@umich.edu

EDUCATION

UNIVERSITY OF MICHIGAN ANN ARBOR

PH.D. ELECTRICAL ENGINEERING
Present | Ann Arbor, Michigan
Graduation Date: May 2026
Cum. GPA: [3.95 / 4.0](#)
Research Advisors:
[Laura Balzano](#), [Jeffrey Fessler](#),
[Scott Peltier](#)

UNIVERSITY OF MICHIGAN ANN ARBOR

M.S. ELECTRICAL ENGINEERING
Spring 2023 | Ann Arbor, Michigan
Cum. GPA: [3.95 / 4.0](#)
Concentration:
AI & Machine Learning
Notable Works:
[Modern Vision Architectures](#)
[Human Pose Estimation](#)
[Matrix Factorization Survey](#)
[Metropolis-Hastings Sampling Method](#)
[Zeroth-Order Optimization Methods](#)
[Deep Learning Projects](#)

UNIVERSITY OF TEXAS ARLINGTON

B.S. ELECTRICAL ENGINEERING
B.S. MATHEMATICS
Spring 2020 | Arlington, Texas
Cum. GPA: [4.0 / 4.0](#)
Capstone Project:
[Wheelchair Dynamometer](#)

Notable Works:
[Machine Learning Algorithms](#)
[Stochastic Models & Simulations](#)
[Numerical Analysis Algorithms](#)
[Image/Video Compression](#)
[WiFi Antenna Optimization](#)

SKILLS

CERTIFICATIONS

[LPI Linux Essentials](#)
[CompTIA A+](#)
[SolidWorks Associate](#)
[AutoCAD Professional](#)
[F.E. Electrical & Computer](#)

LANGUAGES

Python/Pytorch | Julia | $\text{\LaTeX}$ | Linux CLI

INDUSTRY EXPERIENCE

KLA | ALGORITHM ENGINEERING INTERN

May 2024 – August 2024 | Ann Arbor, Michigan

Worked with the reticle/photomask team to group majority image patterns and outliers to augment defect classification systems. [\(example\)](#)

Designed various AI/deep learning clustering and outlier detection algorithms to reduce customer intervention time in reticle development.

OLSSON | ASSISTANT ELECTRICAL ENGINEER

Fall 2020 – Fall 2021 | Fort Worth, Texas

Designed electrical power distribution systems for data center clients and analyzed systems for short circuit analysis, arc flash mitigation, and breaker setting coordination. [\(example\)](#)

RESEARCH EXPERIENCE

GRADUATE WORK | UNIVERSITY OF MICHIGAN

Fall 2021 – Present | Ann Arbor, Michigan

Research Interests:

Signal/Image Processing • Machine Learning • Optimization • Applied Math

Research Topics:

- Deep learning architectures for representation learning of Alzheimer's Disease and behavior score regression in functional MRI
- Heteroscedastic data in ML algorithms such as PCA subspace dimensionality reduction and clustering
- Optimization techniques for inverse problems in low-rank settings for applications such as dynamic MRI

MRI MESH & SLING SEGMENTATION

UT SOUTHWESTERN MEDICAL CENTER | RESEARCH ASSISTANT

Fall 2018 – Summer 2019 | Dallas, Texas

Researched machine learning methods of identification and segmentation of meshes & slings, along with common pelvic floor features, in MR images for radiologist assistance in 3D visualization-guided surgery planning.

[Presented "Automatic Segmentation & 3D Visualization of MR Images of Pelvic Floor"](#) at the [Modern Challenges in Imaging](#) conference located at Tufts University, Medford, Massachusetts.

RECENT PUBLICATIONS

- [1] R. Lobos, J. Salazar Cavazos, R. R. Nadakuditi, and J. A. Fessler, "Smooth optimization algorithms for global and locally low-rank regularizers," *Under review at SIAM Journal on Optimization*, 2025.
- [2] J. Salazar Cavazos, J. A. Fessler, and L. Balzano, "Alpcahus: Subspace clustering for sample-wise heteroscedastic data," *Under review at IEEE Transactions on Signal Processing*, 2025.
- [3] J. Salazar Cavazos and S. Peltier, "Alzheimer's disease classification in functional mri with 4d joint temporal-spatial kernels in novel 4d cnn model," in *International Society for Magnetic Resonance in Medicine, ISMRM*, 2025, p. 3398.
- [4] J. Salazar Cavazos, J. A. Fessler, and L. Balzano, "Alpcah: Subspace learning for sample-wise heteroscedastic data," *IEEE Transactions on Signal Processing*, 2025.